

Career #1

Industry: Cybersecurity/ Network Security

Job: Network Architect

Why it's important: Network architects design and plan high-level and strategic network infrastructure

Career #2

Industry: Web Development

Job: Full Stack Developer

Why it's important:

Career #3

Industry: Artificial intelligence

Job: Machine Learning Engineer

Why it's important: They build the algorithms that are used in the AI process

Network Architect:

What day-to-day looks like:

They help create the plan for data communication networks then they present those designs to the management, customers and staff. Then they deploy said networks and test to ensure there are no bugs in the system while also documenting the process. They also upgrade previous hardware to better models or to make them more efficient. While also researching better ways to do certain things in the work place.

What Industries hire this role:

Educational Services hire network architects including public private or state. They are also hired in telecommunications, temporary help services, Computer systems design and related services and company and enterprise management

Core Skills:

Network architects need to be familiar with Python, Powershell, JSON, Perl and C/C+. To be a network architect you also need analytical skills as you have to think about how to setup a network that best fits the needs and requirements of the company. You also need interpersonal skills because in network architecture you will have to work alongside other people so if you don't work well for you this is no go. You'll also need leadership skills as a lot of network architects direct a team of people to design, build and implement networks.

Educational Pathways:

The degree needed to be a network architect can vary some companies require a bachelors degree. While some require a masters degree. And others show no preference degree or not. The best way to get hired as a network architect without any college degree is to have at least 5 years of experience in IT. While also having a decent amount of industry certifications such as the CCNA, CCNP, and CISSP.

Certifications:

Certification Name	Issuing Organization
Oracle Financials Cloud: Payables 2024 Implementation Professional - Delta	Oracle
Network Automation	OpenText

Tools and Technologies:

They use tools such as VS Code/ Notepad++ for writing code. They also use LucidChart for making diagrams and flowcharts

Salary and Job Outlook:

Entry Level Salary is between \$68,000 - \$90,000 annually

Mid-Level Salary is between \$90,000 - \$130,000 annually

Every year, there are about 11,200 job openings for network architects which is projected to grow by atleast 12% from 2024 to 2034

FullStack Developer:

What the day to day looks like:

They focus several hours of their day on coding, such as new API endpoints, Database Schemas, or implementing business logic. They also help colleagues debug the code using peer review. They also write SQL or deal with NoSQL databases to manage data flow, organization, and security

What Industries hire this role:

They are in high demand in the Tech , Finance, E-Commerce, Military, and Healthcare industries.

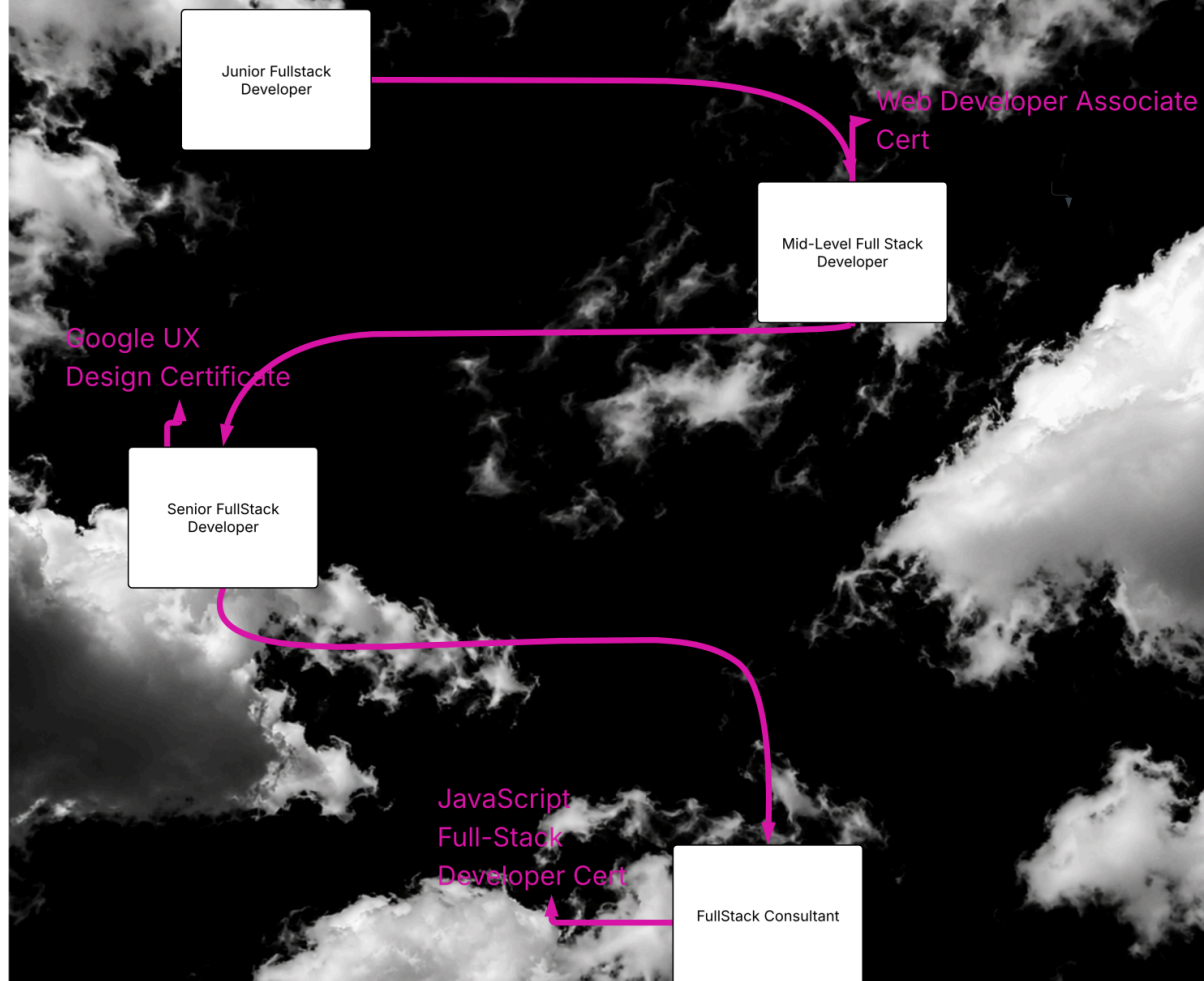
Core Skills:

Those languages you should know are HTML, JavaScript([Node.JS](#)), Python, SQL, Java, C#. Soft skills they need is being able to work in a team with other people for bigger projects and managing pre-existing ones. They also need problem solving skills to fix any issues that come up in or after development.

Education Pathways:

Some companies require backend developers to have a bachelors degree in a specific field but this is not required as some look more for technical skill rather than a degree. A useful bootcamp could be MITxPRO Full Stack Development with MERN bootcamp even though you won't be using all of the things you learn in this bootcamp as a backend developer alot of it can be applied to that specific field.

Career Ladder



<https://www.bls.gov/ooh/computer-and-information-technology/web-developers.htm>

<https://www.bls.gov/ooh/computer-and-information-technology/computer-and-information-research-scientists.htm>

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